LYLE OKOTH

MACHINE LEARNING ENGINEER | SPEECH & AUDIO PROCESSING

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PROFESSIONAL SUMMARY

Machine Learning Engineer with 6+ years of experience designing, training and productionising multilingual speech systems — automatic speech recognition, speech-to-speech translation and text-to-speech synthesis — for East African languages. Owns the full model lifecycle (data, training, evaluation, deployment): halved Swahili/Sheng ASR error rates, shipped real-time speech translation at 310 ms median latency, and cut inference costs 58% through quantisation. Backed by deep software and data engineering roots, and published open datasets and models with 18,000+ downloads on Hugging Face.

CORE COMPETENCIES

Core ML: ASR (Whisper), TTS & voice cloning, speech-to-speech translation, transformer architectures, diffusion & flow-matching models, LLM fine-tuning (LoRA / QLoRA), RLHF & preference optimisation, synthetic data generation, multilingual & low-resource speech

Speech Engineering: Speaker diarisation, VAD, voice embeddings, streaming inference, latency optimisation, time-to-first-token budgets, model evaluation (WER, CER, MOS)

MLOps: Experiment tracking (W&B), model registry, CI regression gates, quantisation (int8), TensorRT/ONNX, GPU serving (CUDA, Triton), Kubernetes, monitoring & model-drift detection

Data & Product: Audio data collection, alignment & annotation platforms, dataset versioning (DVC), data-quality frameworks, research-to-production, technical writing, UI design for speech products

TECHNICAL SKILLS

Languages: Python, C++, JavaScript/TypeScript, SQL

ML Frameworks: PyTorch, Hugging Face Transformers, Diffusers, scikit-learn, LoRA/PEFT, NLTK, spaCy

Serving & Infra: FastAPI, Triton Inference Server, CUDA, TensorRT, ONNX, WebRTC, Redis, Celery

Data & MLOps: Apache Spark, Kafka, Airflow, dbt, DVC, Great Expectations, Weights & Biases, Docker, Kubernetes, Terraform

Cloud: AWS (S3, GPU instances), GCP

PROFESSIONAL EXPERIENCE

Machine Learning Engineer — SautiFlow Labs, Nairobi 2024 – Present

- Cut Swahili and Sheng ASR word error rate from 23.4% to 11.8% by LoRA fine-tuning Whisper-large-v3 on 1,240 hours of in-house audio — the company's first production-grade transcription product for East Africa.
- Architected and shipped streaming speech-to-speech translation at 310 ms median first-audio latency (p95 780 ms) on a single T4 GPU — a 4.5x improvement over the 1.4 s batch baseline.
- Compressed a 1.6 GB TTS checkpoint to 480 MB using int8 quantisation and KV caching, raising capacity to 42 concurrent voices per GPU and cutting inference cost by 58%.
- Built the model evaluation harness (WER, CER, MOS with CI regression gates), blocking 9 model regressions from release across two quarters.
- Published 3 open speech datasets and 2 fine-tuned models to Hugging Face, reaching 18,000+ downloads in the first quarter.

- Built an audio data collection, editing and annotation platform leveraging fine-tuned audio models for AI pre-labelling, reducing human labelling time by 68%.
- Developed speaker diarisation, VAD and voice-embedding pipelines for multi-speaker meeting audio, and RL preference-tuning guardrails against prompt injection and model misuse.
- Set the team's model-card and experiment-tracking standard across 60+ training runs per month; implemented latest speech research to keep models close to state of the art.

Data Engineer — SautiFlow Labs, Nairobi 2022 – 2024

- Designed batch and streaming pipelines (Airflow, Kafka, Spark) processing 6 TB / 38,000 hours of raw audio per year into training-ready speech datasets.
- Automated transcription alignment and diarisation pre-labelling, cutting human labelling from 11 to 3.5 minutes per hour of audio; introduced DVC dataset versioning so every model release traces to an exact data snapshot.

Software Engineer — SautiFlow Labs, Nairobi 2020 – 2022

- Shipped full-stack features for IVR and WhatsApp voice assistants serving 40,000+ monthly users, and trained the team's first production model — a FastAPI-served intent classifier deflecting 28% of support chats.

SELECTED PROJECTS

Vocalis — Brand Voice Studio — Voice-cloning TTS service: one licensed studio read in, a reusable brand voice out, served over an API. Built for African-accented English and local-language narration that vendor catalogs do not cover.

EchoID — Speaker Attribution — Turns unlabelled audio archives into speaker-labelled, searchable transcripts using diarisation, voice embeddings and ASR — queryable by legal, research and support teams.

LabelSet — Speech Data Platform — End-to-end data pipeline behind every model shipped: sourcing recordings, human + AI collaborative labelling, and quality verification before training.

FaceForm — Synthetic Face Generator — Generates consistent, clearly synthetic faces for teams that need people in their visuals without licensing real ones — a designer sets an archetype once, then draws a whole persona family from the same latent noise.

EDUCATION

BSc, Computer Security and Forensics — Jaramogi Oginga Odinga University of Science and Technology (JOOUST) 2016 – 2020

Diploma, Mechatronics Engineering — JKUAT, Juja, Kenya 2013 – 2016

CERTIFICATIONS

DeepLearning.AI, NVIDIA Deep Learning Institute, AWS and Google Cloud professional certifications (2021 – 2025).

REFERENCES

Available on request.